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1. (b) Data and a pointer to the next node.
2. (C) NULL.
3. (b) Self-referential structure.
4. (b) $\text{head} == \text{NULL}$.
5. (b) $\text{newnode} \rightarrow \text{next} = \text{head}; \text{head} = \text{newnode};$
6. (b) $\text{temp} = \text{head}; \text{head} = \text{head} \rightarrow \text{next}; \text{free}(\text{temp});$
7. (C) Traversing from tail to head.
8. (b) n

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9. (C) Copying the data of $p \rightarrow \text{next}$ into p, then deleting $p \rightarrow \text{next}$
10. (b) while ($p \neq \text{NULL}$)
11. (b) $p \rightarrow \text{next} == \text{NULL}$
12. (C) Two pointers advancing at different rates.
13. (C) Three.
14. (b) Exactly one node.
15. (b) $p = (\text{struct node}^*) \text{malloc}(\text{size of } (\text{struct node}));$
16. (b) Avoid special-case handling for insertion and deletion at the head.
17. (C) The first node.
18. (b) It does not support random (indexed) access
19. (b) Loss of access to all nodes, causing a memory leak.
20. (b) A slow pointer and a fast pointer.

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21. (c) The last node.

22. (b) Nodes may be scattered anywhere in memory.

```
23. struct node {  
    int data;  
    struct node * next;  
};
```

24. Given:-

10 → 20 → 30 → 40 → NULL

Output:-

30 50 70

25. ~~The main issue is that head~~

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25. The main issue is that head is passed by value, so an empty list case cannot update the caller's head.

correct code:-

```
void insertEnd (struct node ** head, int x) {  
    struct node * t = (struct node *) malloc (sizeof  
        (struct node));
```

```
    t->data = x;
```

```
    t->next = NULL;
```

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```
if (*head == NULL) {
```

```
    *head = f;
```

```
    return;
```

```
}
```

```
    struct node *p = *head;
```

```
    while (p->next != NULL) {
```

```
        p = p->next;
```

```
        p->next = f;
```

```
}
```

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26. struct node* reverse(struct node* head) {

```
    struct node *prev = NULL;
```

```
    struct node *curr = head;
```

```
    struct node *next = NULL;
```

```
    while (curr != NULL) {
```

```
        next = curr->next;
```

```
        curr->next = prev;
```

```
        prev = curr;
```

```
        curr = next;
```

```
}
```

```
    return prev;
```

```
}
```

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27.

Suppose the List is:-

10 → 20 → 30 → 40 → NULL

To delete 30, first make the previous node (20) point to the node after 30:

Before:

20 → 30 → 40
 ↑
 delete

After:

20 → 40

Pointers adjustment:-

prev → next = temp → next;
free(temp);

Here temp points to the node being deleted.

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28.

Initial List:- 5 → 8 → 12 → 3 → NULL

(i) Insert 7 at beginning

7 → 5 → 8 → 12 → 3 → NULL

(ii) Delete node containing 12

7 → 5 → 8 → 3 → NULL

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(iii) Insert 9 at end

7 → 5 → 8 → 3 → 9 → NULL